Requirement Engineering & Design Concepts

Software Design and Architecture - Question Bank (1,2,3,4 Marks)

PART A: 1 Mark Questions

1 What is Requirement Engineering?
2 Define Requirement Modeling.
3 What is an SRS document?
4 What are functional requirements?
5 What are non-functional requirements?
6 What is a stakeholder?
7 What is requirement elicitation?
8 What is validation in Requirement Engineering?
9 Define Software Design.
10 What is Software Architecture?
11 What is cohesion?
12 What is coupling?
13 What is a Use Case Diagram?
14 What is a Class Diagram?
15 What is a Sequence Diagram?
16 What is an Activity Diagram?

PART B: 2 Marks Questions

1 List the seven tasks of Requirement Engineering.
2 Differentiate between functional and non-functional requirements.
3 What are the main components of an SRS document?
4 Explain requirement validation briefly.
5 Write two characteristics of a good SRS.
6 Define design concepts in software engineering.
7 Explain abstraction and modularity.
8 Differentiate between cohesion and coupling.
9 What is architectural design?
10 Write the purpose of UML diagrams.

PART C: 3 Marks Questions

1 Explain the seven tasks of Requirement Engineering.
2 Describe the process of gathering functional and non-functional requirements.
3 Explain Requirement Modeling with examples.
4 Describe the structure of an SRS document.
5 Explain key design principles in software engineering.

6 Explain architectural design and its importance.
7 Draw and explain a Use Case Diagram with example.
8 Draw and explain a Class Diagram with example.
9 Draw and explain a Sequence Diagram with example.
10 Draw and explain an Activity Diagram with example.

PART D: 4 Marks Questions

1 Explain Requirement Engineering in detail and discuss its seven distinct tasks.
2 What is SRS? Explain its structure and importance in software development.
3 Explain Software Design concepts and principles in detail (abstraction, modularity, cohesion, coupling, etc.).
4 Explain Architectural Design with suitable examples and types of architecture.
5 Explain Use Case, Class, Sequence, and Activity diagrams with examples and neat diagrams.